

# Quantum Computing and Quantum Machine Learning

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## Plans for the week of April 6-104, 2026

- ▶ The Quantum Approximate Optimization Algorithm (QAOA).
- ▶ The HHL algorithm

# What is the QAOA algorithm?

**QAOA** stands for the Quantum Approximate Optimization Algorithm.

Each part of the name reflects something essential about the method:

- ▶ **Quantum:** it runs on a quantum computer, using qubits and quantum gates.
- ▶ **Approximate:** it typically finds near-optimal solutions rather than exact ones.
- ▶ **Optimization:** it is designed to solve optimization problems (like MaxCut, scheduling, portfolio selection).
- ▶ **Algorithm:** it is a well-defined hybrid quantum–classical procedure.

QAOA is a variational quantum algorithm that uses a parameterized quantum circuit to search for good solutions to hard optimization problems.

# What is the HHL algorithm

The HHL algorithm—named after Aram Harrow, Avinatan Hassidim, and Seth Lloyd—is one of the central quantum algorithms for solving linear systems of equations.

The core idea is to solve a linear system

$$Ax = b$$